

NORMAL LIBOR IN ARREARS

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ABSTRACT. We derive the in arrears convexity adjustment in a normal black model.

1. INTRODUCTION

Consider a Libor rate F with fixing time t_f , index start date t_s and index end date t_e which is paid at t_p . We are interested in the valuation of a coupon involving such a Libor rate fixed in arrears, i.e. $t_f = t_p - \Delta$ with Δ denoting the settlement lag (e.g. 2 business days). For the following derivations we assume $\Delta = 0$. We also assume $t_s = t_p =: t$, $t_e = t_s + \tau$ with $\tau = \tau_{\text{accr}}$ the accrual period of the coupon to be valued. These are typically all minor approximations to the reality.

2. THE MODEL

We assume a normal black model for F , i.e.

$$(2.1) \quad dF = \nu dW$$

with ν denoting the normal volatility.

3. THE ADJUSTMENT

The Libor coupon can then be valued (working in the $t + \tau$ forward measure) as

$$(3.1) \quad C = P(0, t + \tau) E^{t+\tau} \left(\frac{F(t)\tau}{P(t, t + \tau)} \right)$$

with $P(t, T)$ denoting the price at time t of a zero bond with maturity T and the expectation being taken in the $t + \tau$ -forward measure $\mathbb{Q}^{t+\tau}$. We note here that our derivation will be done in a single curve setup for the moment, i.e. with identical forwarding and discounting curves.

We write

$$(3.2) \quad F(t) = \frac{F(t)}{1 + F(t)\tau} (1 + F(t)\tau)$$

and get

$$(3.3) \quad C = P(0, t + \tau) E^{t+\tau} (\tau F(t) + \tau^2 F^2(t))$$

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Now $F(\cdot)$ is a $\mathbb{Q}^{t+\tau}$ - martingale. Furthermore by Ito

$$(3.4) \quad dF^2 = 2F dF + \nu dF dF = 2F \nu dW + \nu^2 dt$$

hence

$$(3.5) \quad E^{t+\tau}(F^2(t)) = F^2(0) + \nu^2 t$$

and therefore

$$(3.6) \quad C = P(0, t + \tau)(\tau F(0) + \tau^2 F^2(0) + \tau^2 \nu^2 t)$$

which can be written as

$$(3.7) \quad P(0, t) \tau \left(F(0) + \frac{\tau \nu^2 t}{1 + \tau F(0)} \right)$$

since $P(0, t + \tau) = P(0, t)(1 + \tau F(0))$. It is interesting to note that this adjustment is the same as the in arrears limit case in (3.14) of [1] which is derived in a lognormal model if one relates the normal volatility ν and the lognormal volatility σ by $\nu := F(0)\sigma$, i.e. the drift freezing *approximation* in [1] effectively means to compute the convexity adjustment *exactly* in a normal black model when connecting the parameters of the two models appropriately.

4. CAPLETS

Now we consider an in arrears caplet. Just as above it can be valued

$$(4.1) \quad C = P(0, t + \tau) E^{t+\tau} \left(\frac{(F(t) - K)^+ \tau}{P(t, t + \tau)} \right)$$

We write

$$(4.2) \quad (F(t) - K)^+ = \frac{(F(t) - K)^+}{1 + F(t)\tau} (1 + F(t)\tau)$$

to arrive at

$$(4.3) \quad C = P(0, t + \tau) E^{t+\tau} (\tau(F(t) - K)^+ + \tau^2 F(t)(F(t) - K)^+)$$

We know from above that $F(t)$ is normally distributed with mean $F(0)$ and standard deviation $\nu\sqrt{t}$. Introducing a new variable

$$(4.4) \quad X = \frac{F(t) - F(0)}{\nu\sqrt{t}}$$

which is obviously standard normal we can evaluate the components from the expectation above as

$$(4.5) \quad A := \frac{1}{\sqrt{2\pi}} \int_{(K-F(0))/\nu\sqrt{t}}^{\infty} (X\nu\sqrt{t} + F(0) - K) e^{-X^2/2} dX$$

and

$$(4.6) \quad B := \frac{1}{\sqrt{2\pi}} \int_{(K-F(0))/\nu\sqrt{t}}^{\infty} (X\nu\sqrt{t} + F(0))(X\nu\sqrt{t} + F(0) - K)e^{-X^2/2} dX$$

These integrals can be computed using the general identity

$$(4.7) \quad \frac{1}{\sqrt{2\pi}} \int (aX^2 + bX + c)e^{-X^2/2} dX = \frac{a+c}{2} \operatorname{erf}\left(\frac{X}{\sqrt{2}}\right) - \frac{1}{\sqrt{2\pi}} e^{-X^2/2} (aX + b)$$

and putting all together we retrieve the caplet price as

$$(4.8) \quad C = P(0, t + \tau)(\tau A + \tau^2 B)$$

REFERENCES

- [1] Caspers Peter, *Libor Timing Adjustments*, Electronic copy available at:
<http://ssrn.com/abstract=2170721>
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